BANGLADESH PHYSICS OLYMPIAD 2025

National Round

Category – D

Participant Name:

Registration Number: **Class:**

Name of Institution:

Contact No: **Email:**

Time: 2 Hour

READ THESE INSTRUCTIONS FIRST

- Make sure that Participant Name, Registration number, Class, Name of Institution and Contact number are written clearly.
- Answer all the questions. If there is discrepancy in the two versions use the data provided in the Bengali version.
- Scientific calculator is allowed.

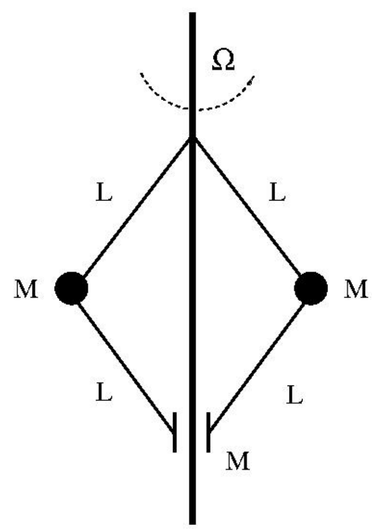
Name of the Examiner:	**Score**
Signature:	

BdPhO-2024-2025, Category D

Feb 28, 2025

1 Spinning Governor

In the old days, steam engines had a failsafe mechanism called a Governor which prevented pressure to develop a certain limit inside the boiler chamber. A scheme of such governer is shown in the accompanying figure where two identical masses, each being m are connected by rods of length L and they are rotated vertically with an angular velocity Ω. There is also another identical mass m which is slide up and down the vertical axis due to its connection to the previous two masses as shown in the figure. Needless to state, the setup in within the Earth's gravitational field.



(a) If the vertical angular deflection of the each of the top rod is θ, what is the coordinate of any of the spinning mass M (shown as black balls in the figure). **2**

(b) What is the kinetic energy of the system, in terms of the angular velocities ? Note that the value of θ is not a fixed value - so that there can contribution from the vertical component of the motion too. (**3**

(c) What is the total energy of the system? **2**

(d) Find the equation of motion for the system. You can use either a vector or scalar method to arrive at your conclusions **2**

2 The Dominant Form of Matter

Most of the matter in the universe remains in the form of plasma, which can loosly speaking -can be thought of as a very light gas but highly ionized. So charges can be though to be free moving. Consider a charge q moving with a velocity $\vec{\mathbf{v}}$ in an electromagnetic field given by the electric field $\vec{\mathbf{E}}$ and an associated magnetic field $\vec{\mathbf{B}}$.

(a) What is the force on the charge (in vector form) when acted on by the above mentioned fields (at the same time)? **1+1**

(b) Think of an observer which is "sitting" on top of the moving charge (we call such observers as *comoving* observers). What will be the force on the charge as measured by this comoving observer? **1**

(c) What would be the relation between the electric and magnetic field as measured by this comoving observer? **1**

(d) However, unlike isolated charges, the charges inside a plasma is surrounded by other charges so this screens the individual charges. To understand the screening, let us think of a spherical shell of charges with an inner radius r and an outer radius $r + dr$. We consider the system to be spherically symmetric. If the electric field on the outer shell is $E_2 = E(r + dr)$ and that on the inner shell is $E_1 = E(r)$, what is net electric flux through the shell? (we take that shell thickness to be so small that methods of differential calculus can be utilized here) **1**

(e) If the electric charge density happens to be ρ what is the total charge inside the shell? **1**

(f) Using Gauss' law for the shell, set up the differential equation for the electric field treating the electric field to be function of r alone. **1**

3 Fitting Elegantly

For a certain fluid, an experimentalist finds the amount of fluid flow V though a pipe is related to the pressure difference X between the pipe ends satisfies a power-law scaling relation: $V = AP^b$ where A and b are constants.

The following data points were obtained :

P	1.1	1.9	3.1	4.05	5.02	7.1	10.5
V	2.05	4.4	7.95	13.1	20.3	40.1	79.3

(a) How should one modify the original relation given above so that the fit can be done in terms of a straight line? **2**

(b)Plot the graph in the provided grid. Label the axes properly and ensure an appropriate scale. **2**

(c) What are the values of A and b that you get from your graph. **2**

(d) Give an estimate in the error for the values you have obtained for A and b. **2**

(e) Can you Interpret the physical meaning of the exponent in this context? **1**

(Graph Grid provided in the next page)